

Randomness Pattern Hunt Student Handout

TOK Cognitive Science Activity Bank • Mathematics • 25-35 min

Category	Mathematics	Time	25-35 min
Difficulty	Easy	ID	randomness-pattern-hunt

Big idea: Human expectations about randomness can mistake clusters for causes and neatness for chance.

Student-Facing Prompt

Inspect Randomness Pattern Hunt Stimulus A. Record one first claim, the evidence you used, your confidence, and what would make you revise.

Concrete Stimulus Packet

Randomness Pattern Hunt Stimulus A	Student-facing stimulus 1: Sequence A: H T H T T H T H H T H T; Sequence B: H H H T T H H T T T H H. Work the example, then decide whether it gives a pattern, a model, a calculation, or a proof.	Students inspect this exact card first and commit to a claim before hearing anyone else.
Randomness Pattern Hunt Stimulus B	Student-facing stimulus 2: Ask: which sequence looks more random, and what makes it feel that way?. Work the example, then decide whether it gives a pattern, a model, a calculation, or a proof.	Use this for comparison, a second round, or a partner challenge.
Randomness Pattern Hunt Reveal / Twist	Student-facing stimulus 3: Ask: which sequence looks more random, and what makes it feel that way?. Work the example, then decide whether it gives a pattern, a model, a calculation, or a proof.	Teacher reveal: connect the changed response to the big idea — Human expectations about randomness can mistake clusters for causes and neatness for chance.

Actual Lesson Questions

#	Question for students
1	After inspecting Randomness Pattern Hunt Stimulus A, what is your first knowledge claim?
2	Which exact detail from Randomness Pattern Hunt Stimulus A did you use as evidence?
3	What did you infer beyond the evidence?
4	How confident are you: 50%, 60%, 70%, 80%, 90%, or 100%?
5	What missing information would most improve the knowledge claim?
6	Why do humans see meaning in random patterns?

Ready-to-Use Examples

- Sequence A: H T H T T H T H H T H T; Sequence B: H H H T T H H T T T H H.
- Ask: which sequence looks more random, and what makes it feel that way?

Student Task

Use the stimulus cards to complete the observation/inference/evidence/confidence grid, then answer one TOK knowledge question connected to Mathematics.

Activity Steps

- 1. Show several short H/T or 0/1 sequences and ask students which look random.
- 2. Students mark streaks, alternations, and suspicious-looking clusters.
- 3. Reveal which sequences were random and which were invented.
- 4. Groups discuss why real randomness often includes streaks.
- 5. Debrief the difference between pattern perception and probabilistic reasoning.

Observation and Evidence Record

Use this grid to keep observation, inference, evidence, and confidence separate.

What I noticed	What I inferred	Evidence	Confidence

TOK Debrief

- Knowledge question: Why do humans see meaning in random patterns?
- When should a pattern make us look for a cause?
- How does mathematics discipline intuition?

Knowledge Questions

- Why do humans see meaning in random patterns?
- When does a pattern justify a causal claim?
- How can mathematics correct intuitive judgement?

Transfer Example

For example, run this activity with a random dot or coin-flip sequence where students hunt for meaningful patterns. Have students commit to an initial claim, then reveal the change, hidden rule, alternative context, or comparison that makes the knowledge problem visible.

Exit Ticket

- One thing I first treated as evidence was...
- My confidence changed because...
- A better knowledge question I can now ask is...